

Test Execution Report

Site Speed Analyzer — Comprehensive Unit & Integration Testing

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Executive Summary



Test Suites: 18/18 passed

Total Duration: 8.30s

Pending: 0

ALL PASSING

Test Framework

Framework	Jest + React Testing Library
Environment	jsdom (jest-environment-jsdom)
Language	TypeScript (ts-jest via next/jest)
Target Application	Site Speed Analyzer (Next.js 16, Static Export)
Test Types	Unit Tests, Component Tests, Integration Tests
Coverage	7 utility modules, 9 UI components, 1 page integration

Test Suite Results

Test Suite	Total	Passed	Failed	Duration	Status
lib/exportCSV	13	13	0	1.52s	PASS
lib/fileParser	17	17	0	0.31s	PASS
lib/reportParser	9	9	0	0.15s	PASS
lib/exportPDF	11	11	0	0.09s	PASS
lib/pagespeedClient	7	7	0	0.08s	PASS
components/ResultsTable	11	11	0	1.07s	PASS
components/ExportDropdown	14	14	0	1.51s	PASS
lib/exportDOCX	7	7	0	0.09s	PASS
lib/urlValidator	17	17	0	0.03s	PASS

Test Suite	Total	Passed	Failed	Duration	Status
components/Tooltip	10	10	0	0.20s	PASS
components/FileUpload	10	10	0	0.56s	PASS
components/URLInput	10	10	0	0.61s	PASS
components/ScoreBadge	13	13	0	0.28s	PASS
components/ScoreChart	6	6	0	0.20s	PASS
smoke	2	2	0	0.01s	PASS
components/MetricsPanel	6	6	0	0.10s	PASS
page	7	7	0	1.43s	PASS
components/LoadingSpinner	9	9	0	0.04s	PASS

Utility Module Tests

Suite	Test Case	Duration	Status
lib/exportCSV	exportCSV (XLS SpreadsheetML) creates a download link and triggers ...	67ms	PASS
lib/exportCSV	exportCSV (XLS SpreadsheetML) creates a Blob with XML content conta...	6ms	PASS
lib/exportCSV	exportCSV (XLS SpreadsheetML) does nothing when reports array is empty	2ms	PASS
lib/exportCSV	exportCSV (XLS SpreadsheetML) does nothing when all reports have em...	1ms	PASS
lib/exportCSV	exportCSV (XLS SpreadsheetML) includes reports with errors (incompl...	3ms	PASS
lib/exportCSV	exportCSV (XLS SpreadsheetML) handles multiple valid reports	13ms	PASS
lib/exportCSV	exportPlainCSV creates a download link and triggers click for valid...	5ms	PASS
lib/exportCSV	exportPlainCSV creates a Blob with CSV content type	6ms	PASS
lib/exportCSV	exportPlainCSV does nothing when reports array is empty	4ms	PASS
lib/exportCSV	exportPlainCSV does nothing when all reports have empty URLs	4ms	PASS
lib/exportCSV	exportPlainCSV includes incomplete reports with blank score fields	4ms	PASS
lib/exportCSV	exportPlainCSV generates correct CSV with BOM and proper headers	1401ms	PASS
lib/exportCSV	exportPlainCSV escapes CSV fields containing commas	5ms	PASS
lib/fileParser	constants exports supported extensions	56ms	PASS
lib/fileParser	constants exports supported MIME types	4ms	PASS
lib/fileParser	parseFile throws for files exceeding 10 MB	103ms	PASS
lib/fileParser	parseFile throws for unsupported file extensions	2ms	PASS
lib/fileParser	parseFile throws for empty files (zero byte arrayBuffer)	3ms	PASS
lib/fileParser	parseFile throws when.xlsx.read fails to parse	4ms	PASS
lib/fileParser	parseFile throws when workbook has no sheets	3ms	PASS
lib/fileParser	parseFile extracts URLs from a sheet with a URL column header	9ms	PASS
lib/fileParser	parseFile scans all cells for URL patterns when no URL header exists	5ms	PASS
lib/fileParser	parseFile throws when no valid URLs are found	5ms	PASS
lib/fileParser	parseFile truncates results beyond 500 URLs and sets truncated flag	104ms	PASS
lib/fileParser	parseFile deduplicates URLs	4ms	PASS
lib/fileParser	parseFile - report detection and resume detects an exported report ...	2ms	PASS
lib/fileParser	parseFile - report detection and resume separates completed and inc...	1ms	PASS
lib/fileParser	parseFile - report detection and resume skips AVERAGE summary rows	2ms	PASS
lib/fileParser	parseFile - report detection and resume defaults strategy to deskto...	2ms	PASS
lib/fileParser	parseFile - report detection and resume throws when report file has...	3ms	PASS

Suite	Test Case	Duration	Status
lib/reportParser	parseReport parses a complete API response correctly	129ms	PASS
lib/reportParser	parseReport returns default report with error when lighthouseResult...	4ms	PASS
lib/reportParser	parseReport returns default report when apiResponse is null/undefined	2ms	PASS
lib/reportParser	parseReport handles missing categories gracefully	3ms	PASS
lib/reportParser	parseReport handles missing audits gracefully	2ms	PASS
lib/reportParser	parseReport handles non-numeric score values	1ms	PASS
lib/reportParser	parseReport handles audits with missing numericValue or displayValue	2ms	PASS
lib/reportParser	parseReport rounds scores correctly	9ms	PASS
lib/reportParser	parseReport handles zero scores correctly	2ms	PASS
lib/exportPDF	exportPDF creates a PDF and calls save with correct filename	52ms	PASS
lib/exportPDF	exportPDF calls autoTable twice (scores table + vitals table)	2ms	PASS
lib/exportPDF	exportPDF does nothing when there are no valid reports	2ms	PASS
lib/exportPDF	exportPDF does nothing when reports array is empty	1ms	PASS
lib/exportPDF	exportPDF renders title and subtitle text	3ms	PASS
lib/exportPDF	exportPDF renders summary score boxes with roundedRect	5ms	PASS
lib/exportPDF	exportPDF draws header bar on first page	3ms	PASS
lib/exportPDF	exportPDF sets up footers on all pages	5ms	PASS
lib/exportPDF	exportPDF handles multiple reports	3ms	PASS
lib/exportPDF	exportPDF passes correct column headers to autoTable	11ms	PASS
lib/exportPDF	exportPDF truncates long URLs in the table body	3ms	PASS
lib/pagespeedClient	fetchReport calls fetch with the correct endpoint and returns data ...	14ms	PASS
lib/pagespeedClient	fetchReport throws on 400 Bad Request with descriptive message	14ms	PASS
lib/pagespeedClient	fetchReport throws on 500 server error	3ms	PASS
lib/pagespeedClient	fetchReport throws on unknown non-OK status	4ms	PASS
lib/pagespeedClient	fetchReport throws a network error when fetch itself throws a TypeE...	8ms	PASS
lib/pagespeedClient	fetchReport throws abort error when external signal is already aborted	17ms	PASS
lib/pagespeedClient	fetchReport retries on 429 with exponential backoff	20ms	PASS
lib/exportDOCX	exportDOCX calls Packer.toBlob and saveAs with correct filename	54ms	PASS
lib/exportDOCX	exportDOCX does nothing when reports array is empty	3ms	PASS
lib/exportDOCX	exportDOCX does nothing when all reports have empty URLs	3ms	PASS
lib/exportDOCX	exportDOCX generates document for reports that include errors (inco...	4ms	PASS
lib/exportDOCX	exportDOCX passes a Document instance to Packer.toBlob	14ms	PASS
lib/exportDOCX	exportDOCX handles multiple reports	7ms	PASS
lib/exportDOCX	exportDOCX handles Packer.toBlob failure gracefully	5ms	PASS
lib/urlValidator	parseURLs splits multiline input into an array of trimmed strings	3ms	PASS
lib/urlValidator	parseURLs handles Windows-style line endings (CRLF)	1ms	PASS
lib/urlValidator	parseURLs trims whitespace from each line	1ms	PASS
lib/urlValidator	parseURLs filters out empty lines	1ms	PASS
lib/urlValidator	parseURLs returns an empty array for empty input	1ms	PASS
lib/urlValidator	parseURLs returns single URL for single-line input	0ms	PASS
lib/urlValidator	validateURL returns true for valid https URLs	3ms	PASS
lib/urlValidator	validateURL returns true for valid http URLs	2ms	PASS
lib/urlValidator	validateURL returns false for ftp or other protocols	3ms	PASS
lib/urlValidator	validateURL returns false for malformed URLs	7ms	PASS

Suite	Test Case	Duration	Status
lib/urlValidator	validateURL returns false for URLs without a protocol	0ms	PASS
lib/urlValidator	validateURL handles URLs with special characters	2ms	PASS
lib/urlValidator	sanitizeAndValidate separates valid and invalid URLs	2ms	PASS
lib/urlValidator	sanitizeAndValidate returns empty arrays for empty input	1ms	PASS
lib/urlValidator	sanitizeAndValidate returns all valid when all URLs are correct	1ms	PASS
lib/urlValidator	sanitizeAndValidate returns all invalid when no URLs are correct	1ms	PASS
lib/urlValidator	sanitizeAndValidate trims whitespace and filters empty lines before...	2ms	PASS

Component Tests

Suite	Test Case	Duration	Status
components/ResultsTable	ResultsTable returns null when results array is empty	108ms	PASS
components/ResultsTable	ResultsTable renders a table with results	403ms	PASS
components/ResultsTable	ResultsTable renders the URL in the table	52ms	PASS
components/ResultsTable	ResultsTable renders score badges for each category	48ms	PASS
components/ResultsTable	ResultsTable shows loading state for reports still being fetched	34ms	PASS
components/ResultsTable	ResultsTable shows error state for failed reports	23ms	PASS
components/ResultsTable	ResultsTable shows success row with expandable details	135ms	PASS
components/ResultsTable	ResultsTable renders column headers	43ms	PASS
components/ResultsTable	ResultsTable renders multiple results	104ms	PASS
components/ResultsTable	ResultsTable shows 'From file' badge for imported results	54ms	PASS
components/ResultsTable	ResultsTable handles mixed loading, error, and success states	70ms	PASS
components/ExportDropdown	ExportDropdown renders the export button	203ms	PASS
components/ExportDropdown	ExportDropdown button has aria-haspopup attribute	355ms	PASS
components/ExportDropdown	ExportDropdown dropdown is closed by default	24ms	PASS
components/ExportDropdown	ExportDropdown opens dropdown on button click	116ms	PASS
components/ExportDropdown	ExportDropdown closes dropdown on second button click	105ms	PASS
components/ExportDropdown	ExportDropdown shows all export options when open	117ms	PASS
components/ExportDropdown	ExportDropdown calls exportCSV when Excel option is clicked	114ms	PASS
components/ExportDropdown	ExportDropdown calls exportPlainCSV when CSV option is clicked	88ms	PASS
components/ExportDropdown	ExportDropdown calls exportPDF when PDF option is clicked	86ms	PASS
components/ExportDropdown	ExportDropdown calls exportDOCX when DOCX option is clicked	68ms	PASS
components/ExportDropdown	ExportDropdown closes dropdown after selecting an export option	62ms	PASS
components/ExportDropdown	ExportDropdown disables button when no completed results exist	47ms	PASS
components/ExportDropdown	ExportDropdown disables button when results have errors	53ms	PASS
components/ExportDropdown	ExportDropdown closes dropdown on Escape key	77ms	PASS
components/Tooltip	Tooltip renders children	27ms	PASS
components/Tooltip	Tooltip does not show tooltip content initially	8ms	PASS
components/Tooltip	Tooltip shows tooltip content on mouse enter after delay	60ms	PASS
components/Tooltip	Tooltip hides tooltip on mouse leave	22ms	PASS
components/Tooltip	Tooltip shows tooltip on focus	15ms	PASS
components/Tooltip	Tooltip hides tooltip on blur	16ms	PASS
components/Tooltip	Tooltip applies custom className to trigger wrapper	7ms	PASS
components/Tooltip	Tooltip trigger has tabIndex=0 for keyboard accessibility	36ms	PASS
components/Tooltip	Infolcon renders an SVG	5ms	PASS

Suite	Test Case	Duration	Status
components/Tooltip	Infolcon has the inline-block class	4ms	PASS
components/FileUpload	FileUpload renders file drop zone with expected text	167ms	PASS
components/FileUpload	FileUpload renders drop zone with correct aria-label	138ms	PASS
components/FileUpload	FileUpload has a hidden file input that accepts csv/xls/xlsx	13ms	PASS
components/FileUpload	FileUpload shows parsing state when processing a file	39ms	PASS
components/FileUpload	FileUpload calls onURLsExtracted when 'Use These URLs' is clicked a...	57ms	PASS
components/FileUpload	FileUpload shows URL count after parsing	31ms	PASS
components/FileUpload	FileUpload shows error state when parseFile throws	28ms	PASS
components/FileUpload	FileUpload resets to idle state when 'Try Again' is clicked after e...	27ms	PASS
components/FileUpload	FileUpload shows imported reports state when parseFile returns reports	41ms	PASS
components/FileUpload	FileUpload renders strategy toggle in idle state	19ms	PASS
components/URLInput	URLInput renders textarea and submit button	245ms	PASS
components/URLInput	URLInput renders with placeholder text	23ms	PASS
components/URLInput	URLInput submit button is disabled when textarea is empty	36ms	PASS
components/URLInput	URLInput submit button is enabled when textarea has content	66ms	PASS
components/URLInput	URLInput calls onSubmit with parsed valid URLs on form submit	50ms	PASS
components/URLInput	URLInput shows invalid URLs when validation fails	52ms	PASS
components/URLInput	URLInput disables textarea and button during loading	52ms	PASS
components/URLInput	URLInput shows 'Generating Reports...' text when loading	18ms	PASS
components/URLInput	URLInput renders StrategyToggle with correct strategy	13ms	PASS
components/URLInput	URLInput does not call onSubmit when only invalid URLs are entered	51ms	PASS
components/ScoreBadge	ScoreBadge renders an SVG element	114ms	PASS
components/ScoreBadge	ScoreBadge renders two circle elements (track + arc)	24ms	PASS
components/ScoreBadge	ScoreBadge renders the score number	15ms	PASS
components/ScoreBadge	ScoreBadge applies green color for score >= 90	30ms	PASS
components/ScoreBadge	ScoreBadge applies amber color for score >= 50 and < 90	9ms	PASS
components/ScoreBadge	ScoreBadge applies red color for score < 50	13ms	PASS
components/ScoreBadge	ScoreBadge clamps score to 0-100 range	9ms	PASS
components/ScoreBadge	ScoreBadge handles score of 0	12ms	PASS
components/ScoreBadge	ScoreBadge renders label when provided	21ms	PASS
components/ScoreBadge	ScoreBadge does not render label when not provided	7ms	PASS
components/ScoreBadge	ScoreBadge renders score at boundary 90 as green	9ms	PASS
components/ScoreBadge	ScoreBadge renders score at boundary 50 as amber	9ms	PASS
components/ScoreBadge	ScoreBadge renders score at boundary 49 as red	9ms	PASS
components/ScoreChart	ScoreChart renders all 4 category labels	107ms	PASS
components/ScoreChart	ScoreChart renders score values as text	22ms	PASS
components/ScoreChart	ScoreChart renders 4 progress bars with correct widths	28ms	PASS
components/ScoreChart	ScoreChart uses green color for scores >= 90	15ms	PASS
components/ScoreChart	ScoreChart uses correct colors for mixed scores	15ms	PASS
components/ScoreChart	ScoreChart renders info icons for tooltips	14ms	PASS
components/MetricsPanel	MetricsPanel renders 5 metric cards	21ms	PASS
components/MetricsPanel	MetricsPanel displays metric display values	19ms	PASS
components/MetricsPanel	MetricsPanel displays full metric names	14ms	PASS
components/MetricsPanel	MetricsPanel renders in a grid layout	13ms	PASS

Suite	Test Case	Duration	Status
components/MetricsPanel	MetricsPanel renders info icons for each metric tooltip	14ms	PASS
components/MetricsPanel	MetricsPanel renders with zero-value metrics without error	23ms	PASS
components/LoadingSpinner	LoadingSpinner renders an SVG element	8ms	PASS
components/LoadingSpinner	LoadingSpinner has aria-label 'Loading'	6ms	PASS
components/LoadingSpinner	LoadingSpinner renders with default medium size	3ms	PASS
components/LoadingSpinner	LoadingSpinner renders with small size	4ms	PASS
components/LoadingSpinner	LoadingSpinner renders with large size	3ms	PASS
components/LoadingSpinner	LoadingSpinner renders two circle elements (track + arc)	2ms	PASS
components/LoadingSpinner	LoadingSpinner has the spinner-rotate class for animation	3ms	PASS
components/LoadingSpinner	LoadingSpinner has correct viewBox for each size	9ms	PASS
components/LoadingSpinner	LoadingSpinner applies inline-block class	3ms	PASS

Integration Tests

Suite	Test Case	Duration	Status
page	Home page renders with dark mode, Desktop strategy, and URL input mode	387ms	PASS
page	Home page switches between URL input and Upload File modes	207ms	PASS
page	Home page toggles between Desktop and Mobile strategy	96ms	PASS
page	Home page toggles dark/light mode on theme button click	86ms	PASS
page	Home page runs analysis: enter URLs, click Generate, see loading, t...	163ms	PASS
page	Home page cancels analysis when Cancel button is clicked	120ms	PASS
page	Home page shows export dropdown and triggers export actions	371ms	PASS

Other

Suite	Test Case	Duration	Status
smoke	Smoke test should verify test infrastructure is working	5ms	PASS
smoke	Smoke test should have access to jsdom environment	2ms	PASS

Test Coverage Summary

Module	Type	Coverage Scope	Status
src/lib/urlValidator.ts	Unit	URL validation, edge cases	PASS
src/lib/reportParser.ts	Unit	API response parsing, score extraction	PASS
src/lib/pagespeedClient.ts	Unit	API calls, retry, timeout, abort	PASS
src/lib/exportCSV.ts	Unit	XLS/CSV generation, formatting	PASS
src/lib/exportPDF.ts	Unit	PDF generation with jsPDF	PASS
src/lib/exportDOCX.ts	Unit	DOCX generation with docx pkg	PASS
src/lib/fileParser.ts	Unit	File parsing, resume detection	PASS
src/components/URLInput.tsx	Component	Rendering, interaction, validation	PASS
src/components/FileUpload.tsx	Component	Upload, drag-drop, parsing	PASS
src/components/ScoreBadge.tsx	Component	SVG ring, color coding	PASS
src/components/ScoreChart.tsx	Component	Bar chart, tooltips	PASS

Module	Type	Coverage Scope	Status
src/components/MetricsPanel.tsx	Component	Metric cards, values	PASS
src/components/Tooltip.tsx	Component	Hover tooltip behavior	PASS
src/components/ResultsTable.tsx	Component	Table/cards, states, sorting	PASS
src/components/ExportDropdown.tsx	Component	Dropdown, export triggers	PASS
src/components/LoadingSpinner.tsx	Component	SVG spinner rendering	PASS
src/app/page.tsx	Integration	Full app flow, state management	PASS

Conclusion

The Site Speed Analyzer application has been thoroughly tested with 179 automated tests across 18 test suites covering all utility modules, UI components, and page-level integration flows. All tests pass with a 100.0% pass rate.

The test suite validates:

- URL validation and sanitization logic
- Google PageSpeed Insights API integration (mocked)
- Report parsing and data transformation
- All export formats (PDF, XLS, CSV, DOCX)
- File upload and resume-from-exported-file parsing
- All UI component rendering and interaction
- Full application flow (analysis, cancel, export)

Build verification: PASSED (npm run build)

Lint verification: PASSED (npm run lint)